

## TUSTIN MCAF, CA, USA

WMO: 722915

Lat: 33.700N

Lon: 117.833W

Elev: 54

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 86-97

WBAN: 93114

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 38.9       | 41.4 | 16.6                          | 12.7 | 57.2 | 21.3 | 16.0 | 59.6 | 25.3                  | 65.8 | 20.3 | 62.7 | 0.6                   | 70   | 0.340 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 19.4                   | 93.1            | 69.7 | 89.4 | 69.1 | 86.3 | 68.5 | 73.4                | 86.2 | 72.1 | 85.0 | 70.9 | 83.4 | 8.0                  | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.0                            | 106.9 | 80.4 | 67.5 | 101.4 | 79.5 | 66.1 | 96.4 | 78.2 | 36.9          | 86.0 | 35.7 | 85.3 | 34.6 | 83.6 | 81.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 16.7          | 13.2 | 11.8 | DB                         | 35.1 | 101.5              | 2.6 | 4.8                                                | 33.2 | 104.9      | 31.7 | 107.7      | 30.2 | 110.4      | 28.3 | 113.8 | (4) |
| (5)               |      |      | WB                         | 31.0 | 76.9               | 2.5 | 2.2                                                | 29.2 | 78.5       | 27.8 | 79.8       | 26.4 | 81.0       | 24.6 | 82.6  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 65.5   | 57.2  | 58.6  | 61.2  | 64.4  | 66.6  | 69.5  | 72.9  | 74.6  | 72.9  | 68.8  | 62.3  | 56.8  |  |  |
|                                                                        | DBStd       | 7.33   | 4.41  | 5.13  | 4.10  | 4.39  | 3.63  | 3.60  | 3.26  | 3.53  | 4.33  | 4.21  | 4.28  | 4.10  |  |  |
|                                                                        | HDD50       | 10     | 3     | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 4     |  |  |
|                                                                        | HDD65       | 1019   | 245   | 185   | 131   | 60    | 25    | 5     | 0     | 0     | 2     | 11    | 101   | 254   |  |  |
|                                                                        | CDD50       | 5675   | 225   | 245   | 347   | 433   | 514   | 585   | 709   | 763   | 686   | 584   | 368   | 216   |  |  |
|                                                                        | CDD65       | 1209   | 2     | 7     | 13    | 43    | 74    | 140   | 244   | 298   | 238   | 130   | 19    | 1     |  |  |
|                                                                        | CDH74       | 9699   | 139   | 179   | 269   | 602   | 456   | 890   | 1546  | 2270  | 1885  | 982   | 413   | 68    |  |  |
|                                                                        | CDH80       | 3064   | 21    | 46    | 77    | 231   | 91    | 238   | 389   | 772   | 710   | 367   | 112   | 10    |  |  |
| (14) Wind                                                              | WSAvg       | 4.1    | 3.4   | 3.9   | 3.8   | 4.6   | 5.0   | 4.9   | 4.9   | 4.5   | 3.9   | 3.7   | 3.3   | 3.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 10.6   | 2.4   | 2.7   | 2.2   | 0.8   | 0.1   | 0.0   | 0.0   | 0.1   | 0.4   | 0.2   | 1.2   | 1.5   |  |  |
|                                                                        | PrecMax     | 26.1   | 7.9   | 12.4  | 10.0  | 4.1   | 0.5   | 0.5   | 0.2   | 1.7   | 1.9   | 1.0   | 5.3   | 4.2   |  |  |
|                                                                        | PrecMin     | 3.5    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 5.9    | 2.3   | 3.0   | 2.3   | 1.1   | 0.1   | 0.1   | 0.0   | 0.4   | 0.6   | 0.2   | 1.2   | 1.3   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 82.9  | 85.8  | 87.7  | 94.9  | 87.3  | 93.3  | 91.6  | 95.1  | 97.4  | 97.4  | 89.5  | 80.8  |  |  |
|                                                                        |             | MCWB   | 57.5  | 61.0  | 63.6  | 64.7  | 69.0  | 69.0  | 71.8  | 71.7  | 71.1  | 67.3  | 63.6  | 57.8  |  |  |
|                                                                        | 2%          | DB     | 78.2  | 79.3  | 80.9  | 86.2  | 82.0  | 85.6  | 87.1  | 90.6  | 91.3  | 89.2  | 82.9  | 75.2  |  |  |
|                                                                        |             | MCWB   | 56.9  | 59.5  | 60.0  | 62.5  | 65.8  | 68.4  | 71.2  | 71.5  | 71.4  | 66.6  | 60.3  | 57.8  |  |  |
|                                                                        | 5%          | DB     | 73.5  | 74.1  | 76.0  | 80.7  | 78.7  | 82.0  | 84.0  | 87.4  | 87.3  | 83.0  | 78.7  | 71.5  |  |  |
|                                                                        |             | MCWB   | 55.8  | 56.9  | 60.0  | 62.9  | 64.4  | 67.5  | 69.9  | 71.1  | 70.7  | 66.8  | 59.2  | 56.8  |  |  |
|                                                                        | 10%         | DB     | 69.0  | 69.8  | 72.0  | 75.8  | 75.9  | 78.8  | 81.5  | 84.1  | 83.7  | 78.7  | 74.7  | 68.0  |  |  |
|                                                                        |             | MCWB   | 54.8  | 56.1  | 60.0  | 62.2  | 63.8  | 66.0  | 69.0  | 70.2  | 69.7  | 65.2  | 59.2  | 56.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 63.6  | 66.4  | 66.2  | 68.0  | 69.7  | 71.6  | 74.6  | 75.3  | 75.2  | 72.6  | 67.5  | 63.8  |  |  |
|                                                                        |             | MCDB   | 72.9  | 80.8  | 78.0  | 82.3  | 82.4  | 86.3  | 86.4  | 89.2  | 89.8  | 85.8  | 77.5  | 73.2  |  |  |
|                                                                        | 2%          | WB     | 61.0  | 62.1  | 63.9  | 66.0  | 67.4  | 69.7  | 72.4  | 73.4  | 73.0  | 70.2  | 65.3  | 61.2  |  |  |
|                                                                        |             | MCDB   | 66.9  | 72.4  | 75.4  | 80.5  | 78.9  | 83.2  | 84.2  | 86.4  | 86.4  | 82.8  | 75.6  | 68.8  |  |  |
|                                                                        | 5%          | WB     | 59.3  | 60.1  | 62.1  | 64.4  | 65.7  | 68.3  | 70.8  | 72.1  | 71.4  | 68.4  | 63.4  | 59.3  |  |  |
|                                                                        |             | MCDB   | 66.5  | 67.8  | 72.7  | 77.7  | 76.7  | 80.3  | 82.1  | 85.1  | 84.6  | 79.1  | 73.5  | 66.5  |  |  |
|                                                                        | 10%         | WB     | 57.5  | 58.4  | 60.4  | 62.7  | 64.3  | 66.8  | 69.5  | 70.7  | 70.1  | 66.7  | 61.6  | 57.6  |  |  |
|                                                                        |             | MCDB   | 65.5  | 65.8  | 70.0  | 74.7  | 74.5  | 77.8  | 80.0  | 83.0  | 82.7  | 76.4  | 71.0  | 65.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 23.4  | 21.4  | 21.0  | 21.7  | 18.1  | 18.1  | 17.7  | 19.4  | 21.3  | 21.7  | 25.2  | 23.5  |  |  |
|                                                                        |             | MCDBR  | 33.9  | 31.4  | 31.2  | 33.5  | 24.3  | 24.9  | 21.5  | 24.6  | 28.2  | 31.3  | 34.0  | 30.7  |  |  |
|                                                                        |             | MCWBR  | 18.3  | 16.8  | 17.1  | 17.0  | 12.6  | 12.6  | 10.6  | 12.0  | 13.5  | 16.2  | 17.9  | 18.2  |  |  |
|                                                                        |             | MCDBR  | 22.4  | 22.4  | 24.3  | 25.2  | 19.9  | 21.6  | 19.4  | 21.3  | 24.2  | 24.2  | 25.4  | 23.7  |  |  |
|                                                                        | 5% WB       | MCWBR  | 14.9  | 14.2  | 15.2  | 14.6  | 11.0  | 11.7  | 10.4  | 11.3  | 12.5  | 13.6  | 16.1  | 16.6  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.323 | 0.332 | 0.356 | 0.363 | 0.361 | 0.376 | 0.383 | 0.367 | 0.353 | 0.353 | 0.311 | 0.310 |  |  |
|                                                                        | taud        |        | 2.536 | 2.523 | 2.430 | 2.418 | 2.457 | 2.416 | 2.467 | 2.495 | 2.526 | 2.500 | 2.610 | 2.613 |  |  |
|                                                                        | Ebn at Noon |        | 283   | 292   | 292   | 293   | 293   | 287   | 285   | 288   | 289   | 280   | 286   | 283   |  |  |
|                                                                        | Edh at Noon |        | 26    | 29    | 35    | 37    | 36    | 37    | 35    | 34    | 31    | 30    | 24    | 23    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1017  | 1296  | 1731  | 2102  | 2253  | 2371  | 2362  | 2249  | 1901  | 1444  | 1124  | 920   |  |  |
|                                                                        | RadStd      |        | 70    | 126   | 91    | 107   | 151   | 146   | 122   | 62    | 75    | 104   | 67    | 68    |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (50 neighbors) | +1.13 | N/A     | -3.06  | +1.30   | N/A   | N/A   | N/A         | -202  | +401  | +202  |

Nomenclature: See separate page